

ADITYA KUMAR SAH

AI & DATA SCIENCE ENGINEER · OPEN-SOURCE CONTRIBUTOR

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PROFILE

Final-year B.Tech (CS + Data Science) student with production experience building memory-augmented AI systems, cloud-native backends, and open-source developer tooling. Proficient in Python, FastAPI, and GCP. Strong interest in scalable systems, LLM applications, and applied ML — actively contributing to the Python open-source ecosystem via a GSoC 2026 submission.

EDUCATION

Deenbandhu Chhotu Ram University of Science & Technology, Murthal

Aug 2023 – May 2027

B.Tech — Computer Science with Data Science Specialization

- Relevant coursework: Data Structures & Algorithms, Machine Learning, Data Analytics, Mobile App Development, Software Engineering, Disaster Management, Professional Communication

EXPERIENCE

Software Engineer Intern — PHP & Full-Stack

Apr 2024 – Jun 2024

121Page.Com · Delhi, India

- Engineered and shipped **production-grade backend modules** in PHP and MySQL covering the full CRUD lifecycle, including user authentication, session management, and role-based access control.
- Identified and resolved critical bugs across staging and production environments, improving application stability and enabling smooth nightly deployments.
- Collaborated with cross-functional teammates in an Agile sprint cycle — delivered all assigned features on schedule and authored onboarding documentation for the backend codebase.

Google Cloud Arcade — Legend Tier Achiever

Jan 2025 – Jun 2025

Google · Highest achievable tier in the programme

- Attained **Legend tier — the highest rank** in the 6-month Google Cloud Arcade programme by completing advanced hands-on labs across Compute Engine, BigQuery, Kubernetes Engine, Cloud Functions, Pub/Sub, and IAM.
- Deployed containerised microservices on GKE and built multi-table BigQuery analytical pipelines, demonstrating end-to-end production cloud architecture skills.
- Demonstrated competency across all four core GCP domains: compute, storage, networking, and identity & access management.

PROJECTS

AudienceIQ — Memory-Augmented AI Social Media Analytics Agent

2025

Python · FastAPI · ChromaDB · SQLite · Groq API (qwen3-32b) · React

- Architected a multi-agent analytics system using a **Hindsight memory loop** — persisting semantic campaign insights in ChromaDB (vector store) + SQLite, enabling context-aware recall and iterative content improvement across sessions.
- Built a FastAPI backend that orchestrates **LLM inference via Groq API** to generate, score, and iteratively refine social media posts; system supports multiple independent learning loops with full audit trails logged per iteration.
- Designed a React dashboard with a real-time memory-insights panel, giving marketers visibility into how content reasoning evolved across successive generation cycles.

Python Dependency Analyzer CLI — GSoC 2026 Proposal · Python Software Foundation

2026

Python · AST Parsing · NetworkX · Click · PyPI Advisory API

- Designed and submitted a full GSoC proposal to the **Python Software Foundation** for an open-source CLI tool that analyzes Python project dependency graphs, detects circular imports, and surfaces known CVEs via the PyPI advisory API.
- Architected a **plugin-extensible AST import scanner** that builds dependency graphs with NetworkX and renders them as JSON/HTML/SVG artefacts — enabling automated visualisation of transitive dependency chains.
- Proposed a compatibility-matrix generator that cross-validates installed package versions against declared Python version constraints, designed to integrate directly into CI pipelines.

[Node.js](#) · [React](#) · [REST APIs](#) · [OAuth 2.0](#)

- ▶ Built an end-to-end web platform enabling brands to track cross-platform engagement metrics and receive AI-powered content recommendations, reducing manual content-planning effort.
- ▶ Implemented **OAuth 2.0 authentication** and a modular RESTful API layer with structured request routing; frontend built in React with component-level state management and live metric dashboards.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, PHP, Bash

Frameworks & Libraries: FastAPI, React, Node.js, scikit-learn, pandas, NumPy, ChromaDB, NetworkX

Cloud & DevOps: Google Cloud Platform (Compute Engine, BigQuery, GKE, Pub/Sub, IAM), AWS, Azure, Docker

Databases: MySQL, SQLite, ChromaDB (vector store)

AI / ML: LLMs, Prompt Engineering, RAG, Vector Databases, Generative AI, Machine Learning, MLOps

Tools & Practices: Git, REST APIs, Agile/Scrum, Zapier, Make (Integromat), A/B Testing, Open-Source Contribution

CERTIFICATIONS & ACHIEVEMENTS

- ▶ **Cloud Computing** — Indian Institute of Technology (IIT) Kharagpur · **Certified**
- ▶ **Data Scientist Certification** — End-to-end ML pipeline development · **Certified**
- ▶ **Google Cloud Arcade Legend Tier** — Highest achievable rank across the 6-month, 50+ lab GCP programme (Jan – Jun 2025)
- ▶ **GSoC 2026 — Python Software Foundation:** Proposal submitted for Python Dependency Analyzer CLI Tool; under review by PSF mentors
- ▶ **Accenture Developer Job Simulation — Forage (May 2024):** Completed practical tasks in cloud migration architecture, IAM policies, SDLC security, algorithm debugging, unit testing, UAT, and technical requirements definition